

• 18.06 Linear Algebra - Lectures 31 & 32
Textbook Notes - Sections 8.2 & 8.3

• Section 8.2: Matrix of a Linear Transform.

It has no single matrix $\rightarrow$ one matrix per choice of bases.
The construction is mechanical so the real interest is which bases make it simplest to compute.

• The Construction Rule

Choose bases $v_1 \dots v_n$ for V and $w_1 \dots w_m$ for W . Then:

[Column j of $A = T(v_j)$ written in the output basis]

So $T(v_j) = a_{1j}w_1 + \dots + a_{mj}w_m$, and these a_{ij} form column j . The matrix is $m \times n$.

Why it works: $v = c_1v_1 + \dots + c_nv_n$, so linearity gives $T(v) = c_1T(v_1) + \dots$
• Matrix-times-vector combines columns identically.
• Get the basis vectors right and linearity does the rest.

• Change of Basis

For $T(v) = v$ with different input and output bases, the matrix is not I : $B = W^{-1} \cdot V$

$\hookrightarrow V$ - holds the input basis $\hookrightarrow W$ - holds the output basis

$\hookrightarrow$ From $Vc = Wd$, so $d = W^{-1}Vc$

Note the reversal: changing from standard basis ($V = I$) to basis W gives $B = W^{-1}$, not W . Larger basis vectors $\rightarrow$ smaller coefficients.

• Composition = Matrix Multiplication

Why matrix multiplication is defined as it is:

$$\left[(TS)(u) = T(S(u)) \text{ matches } (AB)x = A(Bx) \right]$$

- $S: U \rightarrow V$ with matrix B , $T: V \rightarrow W$ with matrix A , so TS has matrix AB and $(m \times n)(n \times p) = (m \times p)$.
- The basis for V must match as it is S 's output & T 's input.

Consequence: rotating by θ twice gives 2θ .
Squaring the rotation matrix yields the double-angle formulas

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

• Choosing the best Basis

Eigenvectors ($V = W$, square).

With n independent eigenvectors, use them both as bases.

↳ the matrix becomes A .

For any other basis in the columns of B :

$$A_{\text{new}} = B^{-1} A B$$

↳ Similar matrices again. Applied to $I \cdot T \cdot I$

• Singular vectors (different V and W , any shape).

↳ Input Basis = the V 's ↳ Output Basis = the U 's

$$B_{\text{out}}^{-1} \cdot A \cdot B_{\text{in}} = U^{-1} A V = \Sigma$$

↳ Not similar, two different bases. Both orthonormal, so they preserve length.

↳ Strang calls this Isometric: $C = Q_1 A Q_2$ is isometric to A for orthogonal Q_1, Q_2 .

• Worked Example

Derivative $T(v) = \frac{dv}{dx}$, input basis $1, x, x^2, x^3$,
output basis $1, x, x^2$:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

Column 1 is zero since $T(1) = 0$.

↳ The kernel of T is the nullspace of A .

↳ Reverse both bases and A becomes diagonal with
 $\sigma = 3, 2, 1$.

The integral gives A^+ , with $AA^+ = I$ but $A^+A \neq I$.

↳ differentiating then integrating loses the constant

↳ the Pseudoinverse

◦ Section 7.4 : The Geometry of the SVD

$A = U\Sigma V^T$ reads as rotate, stretch, rotate.

That picture yields the matrix norm, the polar decomposition, and the pseudoinverse.

• The Geometric Picture

$$A = U\Sigma V^T = (\text{rotation}) (\text{stretching}) (\text{rotation})$$

• A takes the unit circle to an ellipse:

↳ the V_i are radius vectors, mapping to semi-axes $\sigma_i U_i$

↳ For 2×2 , the entries a, b, c, d correspond to $\theta, \sigma_1, \sigma_2, \phi$.

• The Matrix Norm

$$\|A\| = \max (\|Ax\| / \|x\|) = \sigma_1$$

• The largest singular value is the maximum growth factor, achieved at $x = V_1$.

• Two properties follow from $\|Ax\| \leq \|A\| \|x\|$ by dividing and maximising:

$$1. \|A + B\| \leq \|A\| + \|B\| \quad (\text{triangle})$$

$$2. \|AB\| \leq \|A\| \|B\| \quad (\text{product})$$

• Eigenvalues vs Singular Values:

• $|\lambda|_{\max} \leq \sigma_1$ always.

• But $|\lambda_2| \leq \sigma_2$ is false as $|\lambda_1, \lambda_2| = \sigma_1, \sigma_2$ for 2x2,
• $|\lambda_1| \leq \sigma_1$ forces $|\lambda_2| \geq \sigma_2$.

• Eckart-Young-Mirsky

$$\|A - B\| \geq \|A - A_k\| = \sigma_{k+1} \text{ for every } B \text{ of rank } k.$$

↳ with $A_k = \sigma_1 u_1 v_1^T + \dots + \sigma_k u_k v_k^T$

• Truncating the SVD is the provably the best low-rank approximation.

• Corollary: the nearest singular matrix comes from setting $\sigma_{\min}$ to zero, so $\sigma_{\min}$ is the distance from A to singularity.

• Polar Decomposition

$$A = QS$$

• $Q = UV^T$ orthogonal (rotation), $S = V\Sigma V^T$ symmetric positive ~~definiteness~~ semidefinite (stretch).

• Insert $V^T V = I$ into the middle of the SVD

↳ $S^2 = V\Sigma^2 V^T = A^T A$, so S is the positive definite square root of $A^T A$.

↳ eigenvalues of S = singular values of A ; eigenvectors = the V 's

↳ reverse order $A = KQ$ with $K = U\Sigma U^T$, the square root of AA^T

↳ $Q = UV^T$ is the nearest orthogonal matrix to A

• Left & Right Inverses

Three cases by rank, before the general tool:

- $r = m = n$ two-sided A^{-1} $AA^{-1} = A^{-1}A = I$
- $r = n < m$ left $A_l^+ = (A^T A)^{-1} A^T$ $A_l^+ A = I$
- $r = m < n$ right $A_r^+ = A^T (A A^T)^{-1}$ $A A_r^+ = I$
- $r = m, n$ neither $\rightarrow$ pseudoinverse only.

- Full column rank makes $A^T A$ invertible, same fact behind Least Squares in Sec 4.3.
- Full row rank makes $A A^T$ invertible, so $Ax = b$ always has solutions (infinitely many when $n > m$).

• The Pseudoinverse

$$\underline{A^+} = V \Sigma^+ U^T \quad (n \times m)$$

- Σ^+ inverts each nonzero σ and leaves zero rows & cols as 0.

• Its action on the bases:

$$\begin{aligned} \hookrightarrow A^+ u_i &= \frac{v_i}{\sigma_i} & \text{for } i \leq r & \text{ (col space } \rightarrow \text{ row space)} \\ \hookrightarrow A^+ u_i &= 0 & \text{for } i > r & \text{ (left nullspace killed)} \end{aligned}$$

- A sends row space to column space. A^+ sends it back.
- Both A and A^+ have dimension $\leq r$.

• Defining Property: A^+ makes both products projections.

$$\hookrightarrow AA^+ = \text{projection onto } \mathcal{C}(A)$$

$$\hookrightarrow A^+A = \text{projection onto } \mathcal{C}(A^T)$$

- When A^{-1} exists, $A^+ = A^{-1}$. For rank 1 $A = \sigma uv^T$, pseudoinverse: $A^+ = \frac{v u^T}{\sigma}$

• Least Squares with Dependent Columns

When A has dependent columns, $A^T A$ is singular and $A^T A \hat{x} = A^T b$ has infinitely many solutions.

• The pseudoinverse picks one:

$$x^+ = A^+ b$$

This is the shortest solution where the solution is in the row space with nullspace components removed.

• Any other solution is x^+ plus a nullspace vector, which adds length (by Pythagoras).

→ Completes Chapter 4: least squares handles b outside the column space, the pseudoinverse handles dependent columns.